

10-22-25 Research Review

July – October 2025 for Integer
Previous presentation on 07-02-25

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The Idea

Many AI models exist to detect objects, but few of them are trained for small maritime objects.

Training AI to identify marine objects requires labeling thousands of images, a tedious and costly process



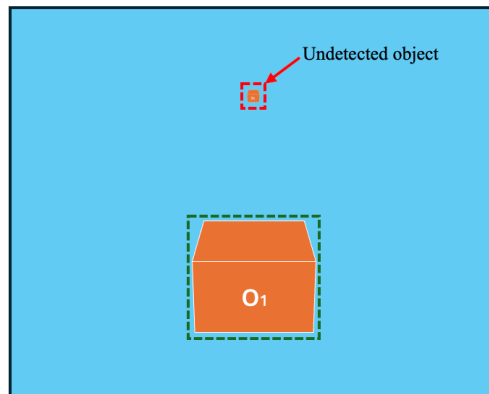
Why Radar?

- **Camera**

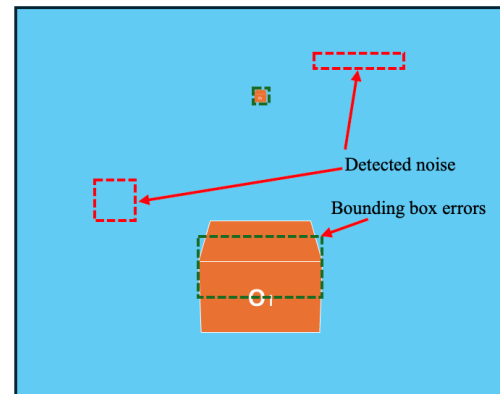
- Gives us a high-resolution, color picture of the world.
- Great for identifying what an object is (a boat, a buoy, etc.).
- Doesn't work well in fog, rain, or at night.

- **Radar**

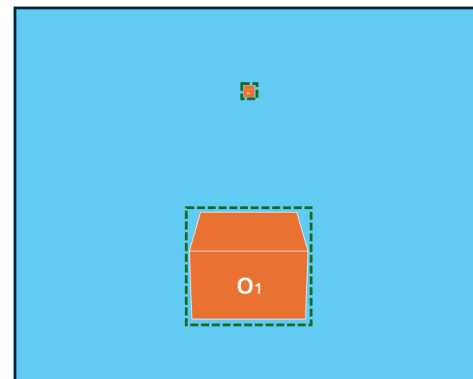
- Sends out radio waves and listens for echoes.
- Excellent at detecting objects at long distances, day or night, in any weather.
- Gives a low-resolution, "blob-like" picture.



(a) Vision only



(b) Radar only



(c) Sensor Fusion

Auto-labelling

The radar scans the area and returns a sweep

Detection Team calculates the object's location on sweep

Rectification Team converts radar coordinates into locations on camera

Draw a bounding box on the camera image where object is detected

Goal

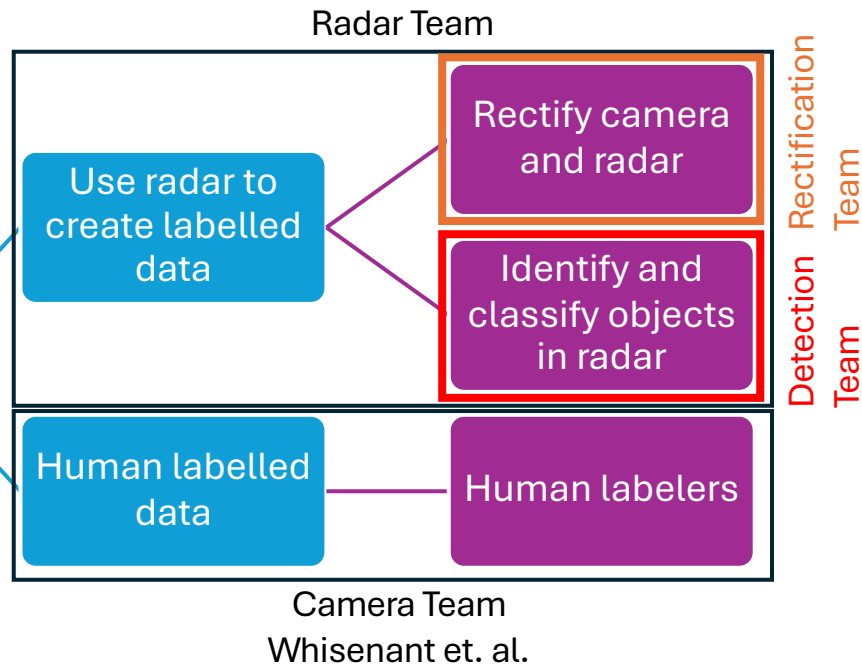
Develop model
to identify small
objects in ocean

Sub Goal

Build model
architecture

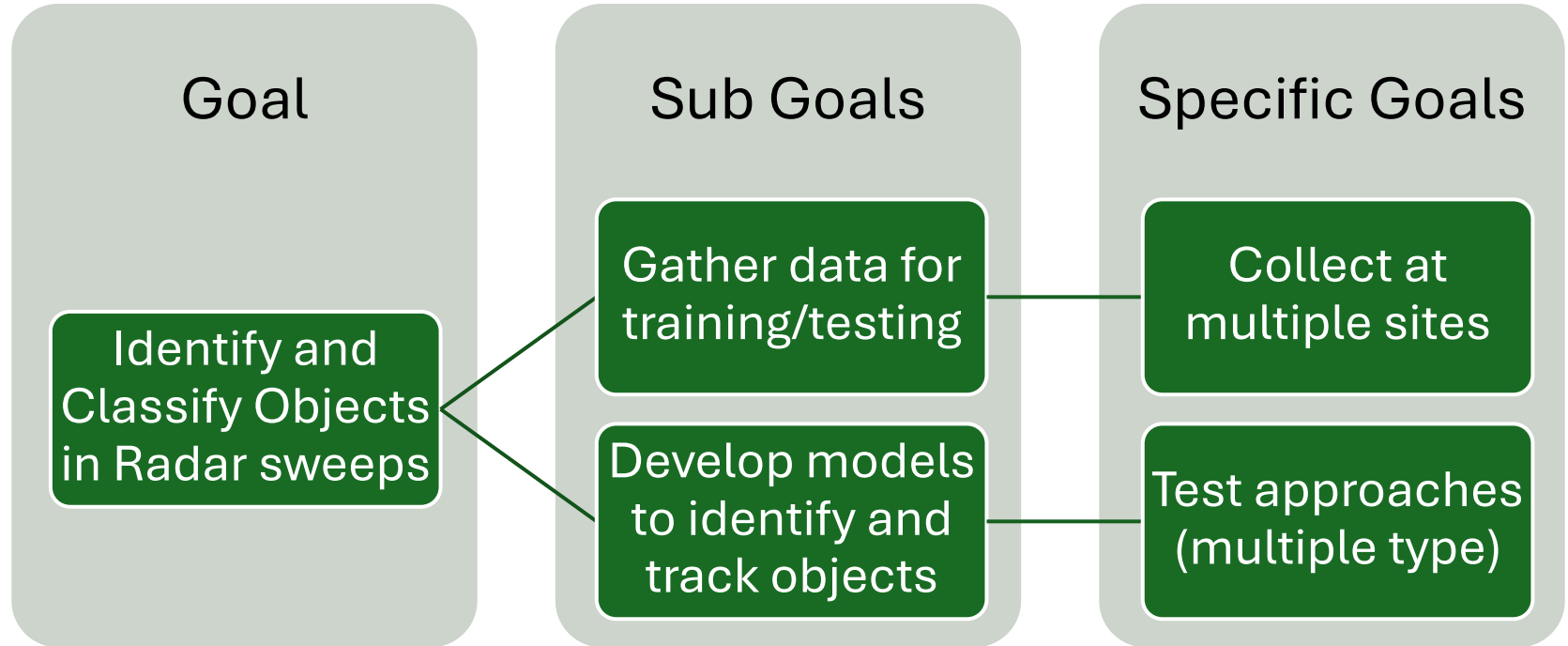
Gather labelled
data for model
training

Overview of Project

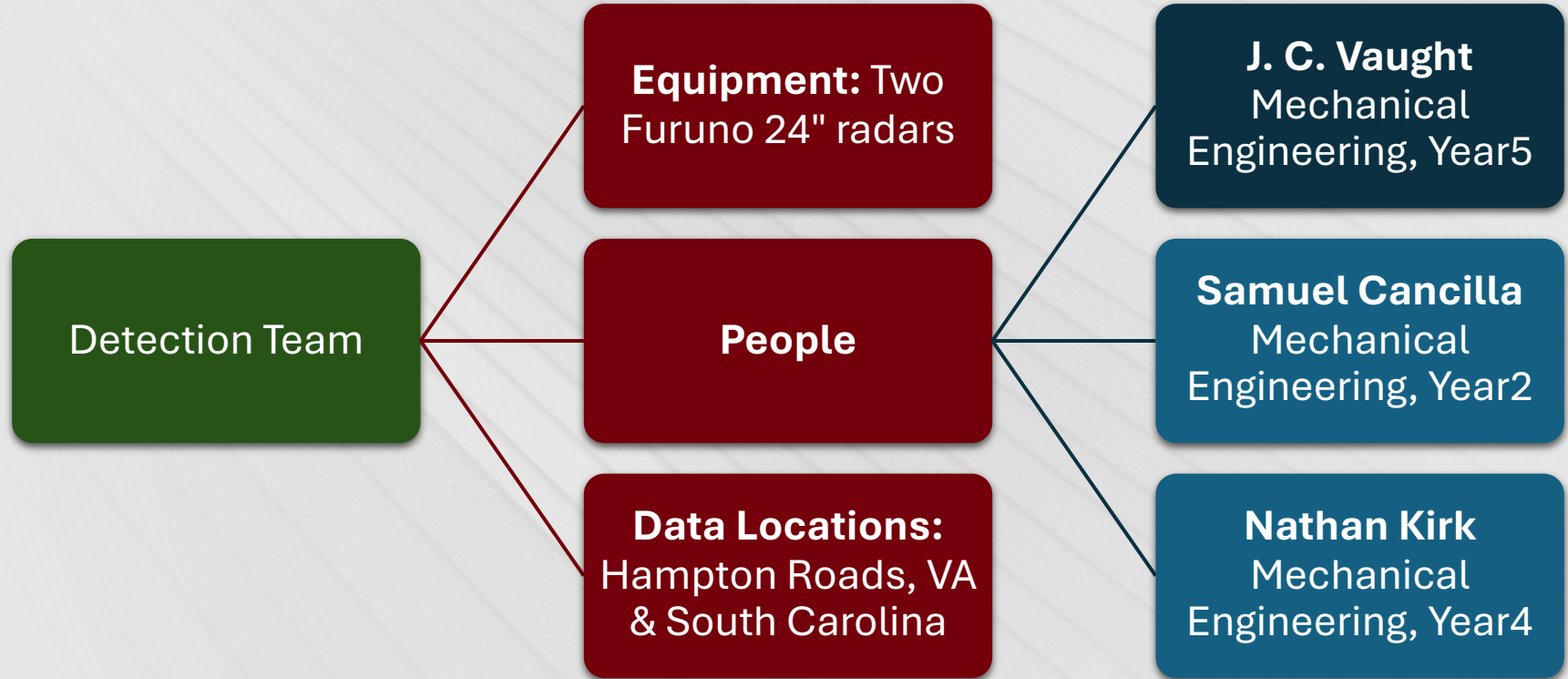


Whisenant et. al.

Detection Team Goals



Team & Equipment



Traditional Pipeline

Collect
Data



Land v
Object

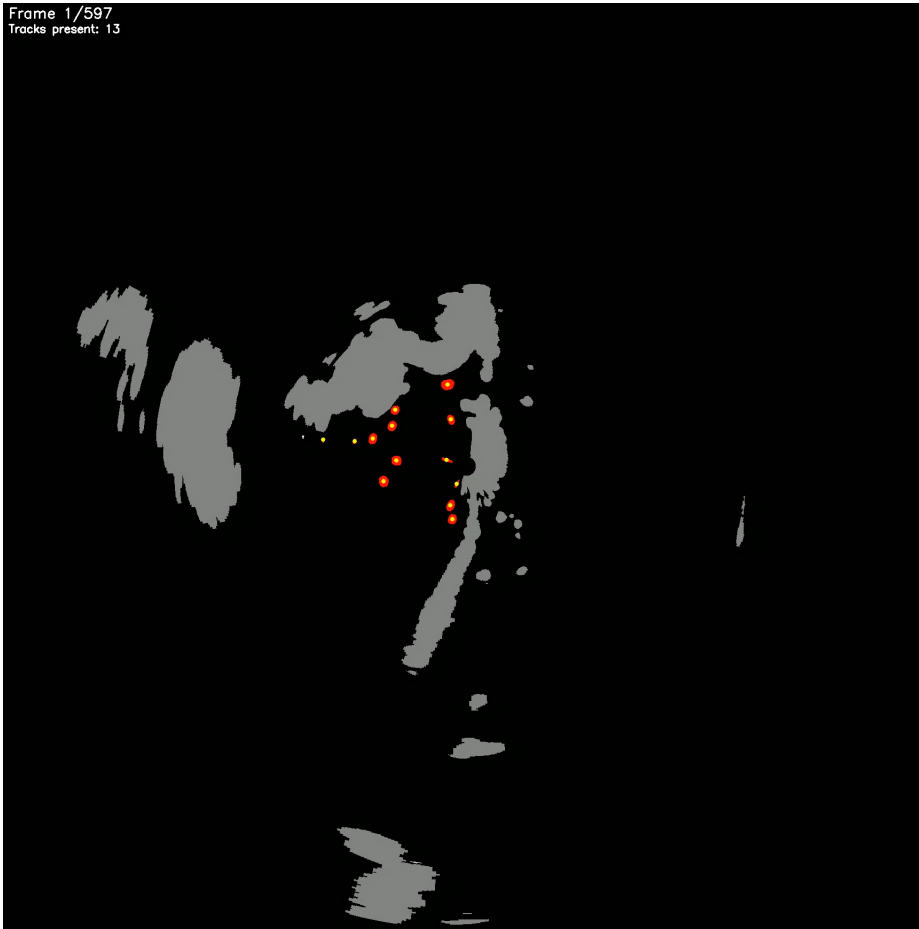


Moving v
Stationary



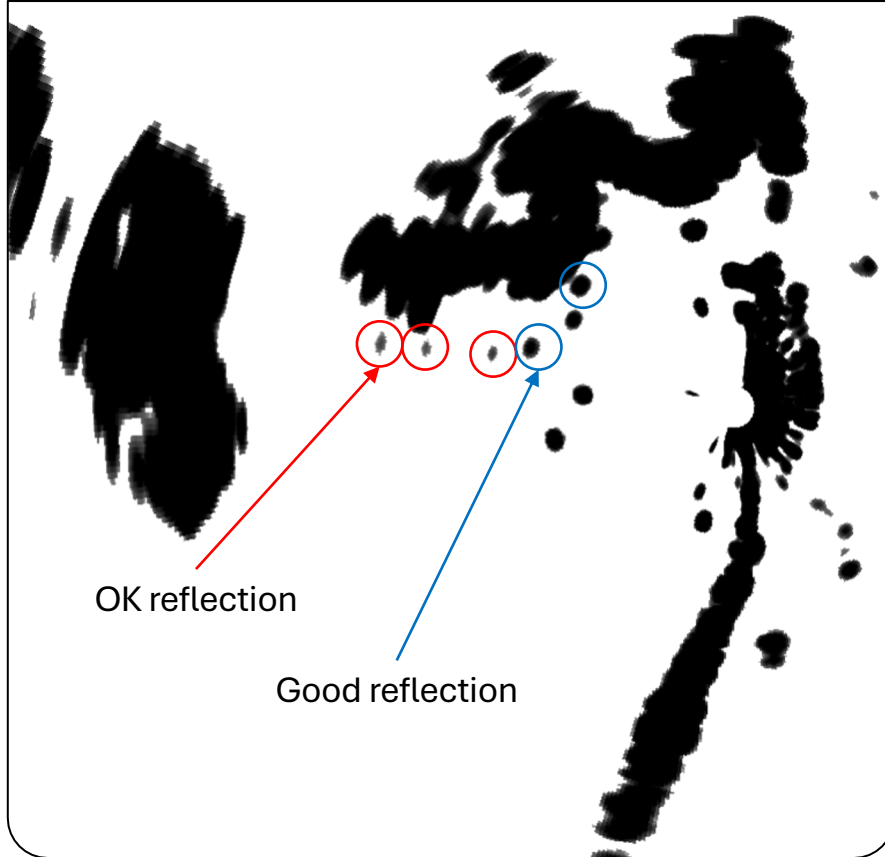
Tracking

Frame 1/597
Tracks present: 13

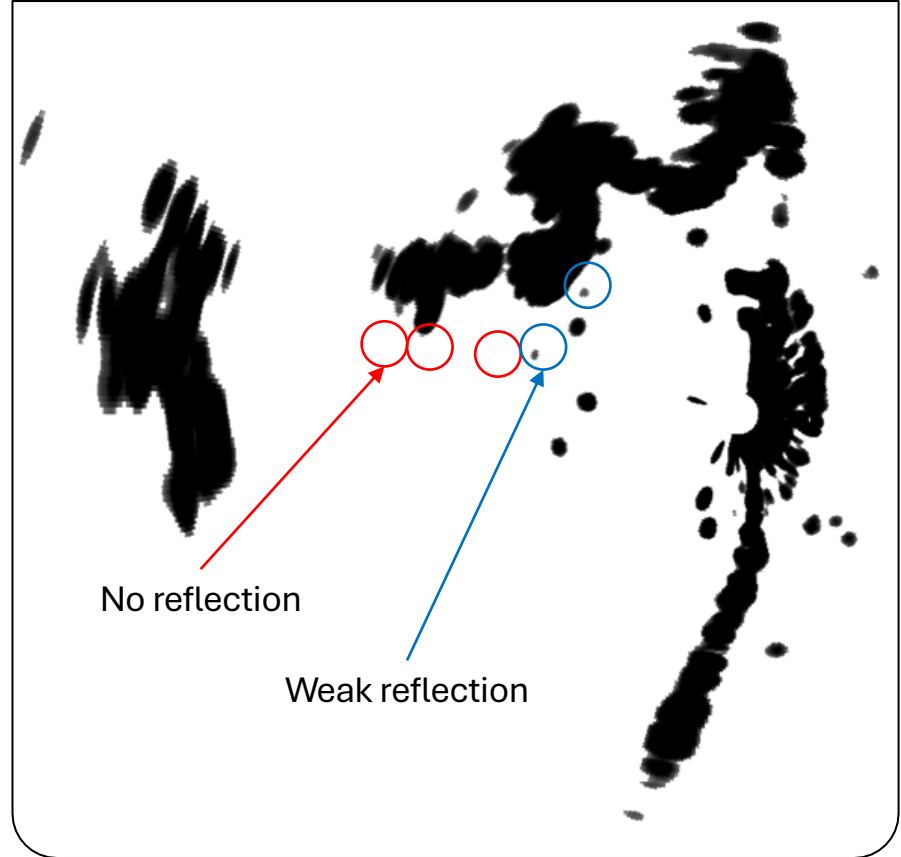


So why is this hard/novel?

'Good' Radar Sweep



Bad Radar Sweep



Seems easy – just do ...

- Most tracking/classification software is built for RGB camera tracking
- Objects don't tend to disappear and reappear in those frames at random
- Some possible solutions:
 - Use higher gain? This also increase the p_{fa} and noise rate
 - Use multiple gains? Doesn't work for fast-moving boats
 - Use longer pulses? We get reduced object position resolution
- Our solution is twofold:
 - a) Combine multiple gains & pulse lengths into an HDR-style algorithm before running through standard tracking
 - b) Build a tracking algorithm from ground-up to account for missing frames more accurately

Completed Work

HDR, Pulses, Entropy, MAE & ML

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Completed work summary (July – October)

July 2025:

- HDR gain comparison
- Radar entropy analysis research
- Radar calibration and radar simulation work

August 2025:

- Object detection methodology development
- Object tracking integration with MAE-ML architecture

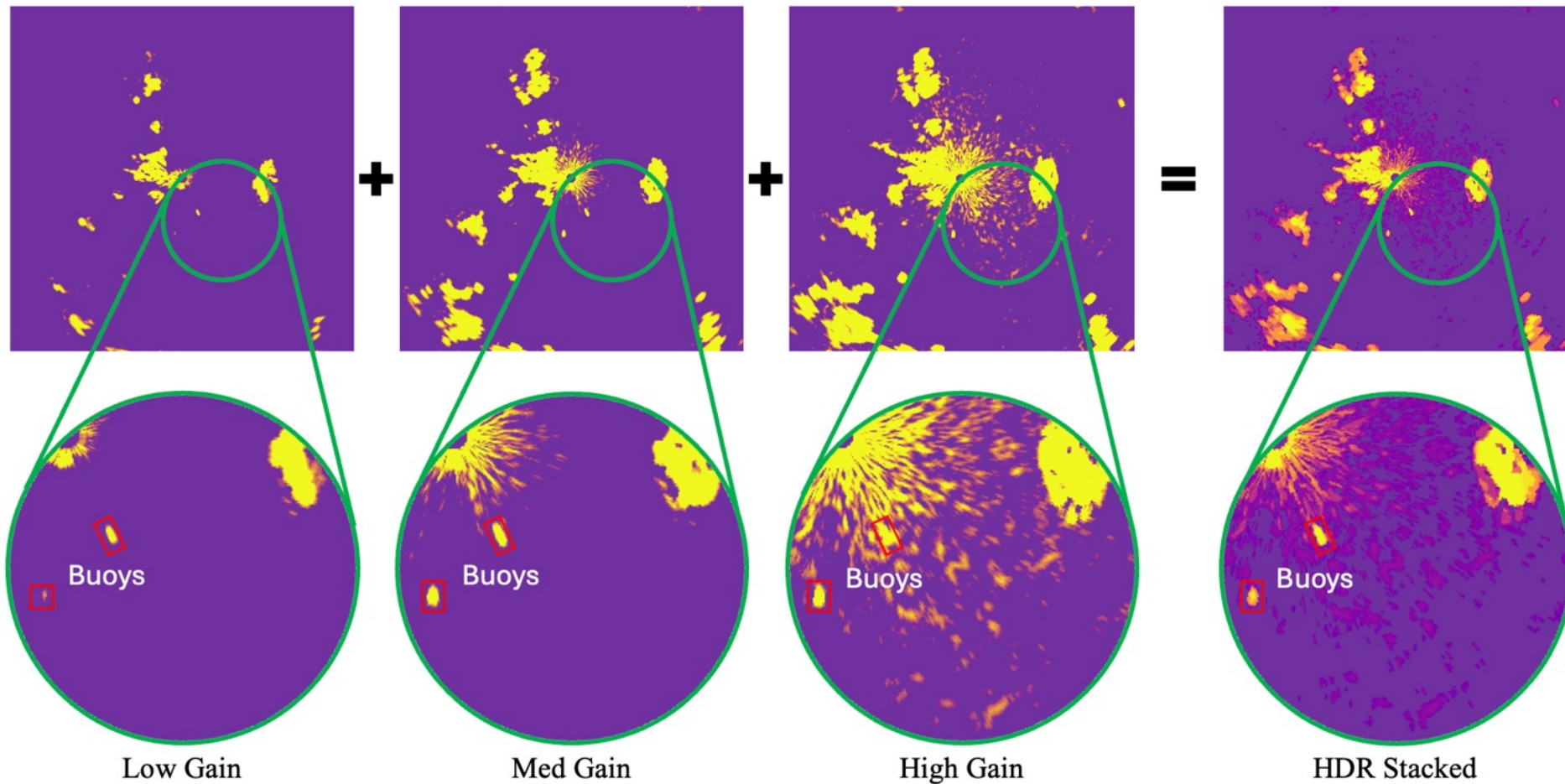
September 2025:

- MAE architecture sliding window improvements
- GPU sharding implementation & libraries

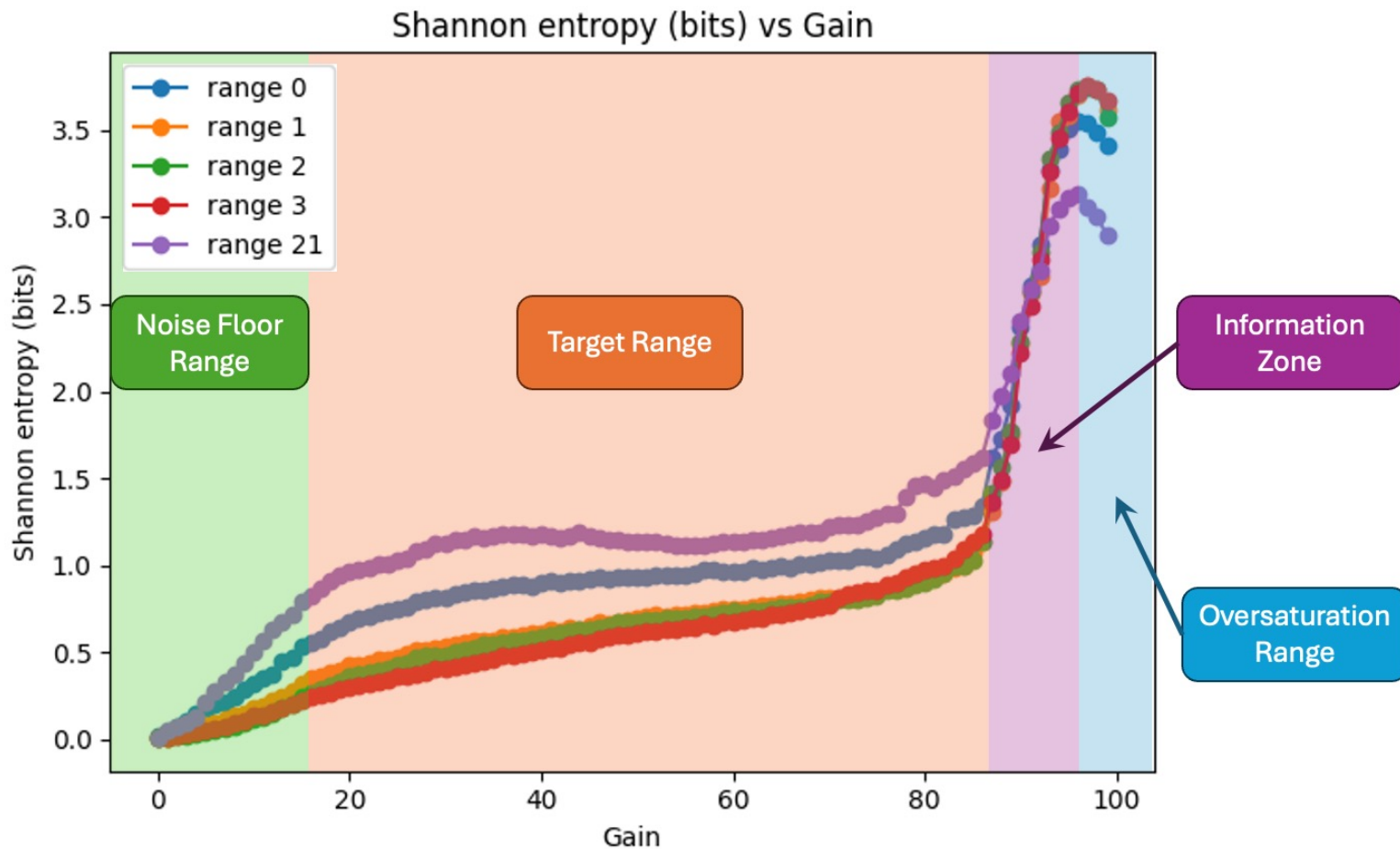
October 2025:

- Literature review on tracking methods for radar
- Literature review on annotation methods for images & radar

HDR Gain – July '25



Radar Entropy – July '25



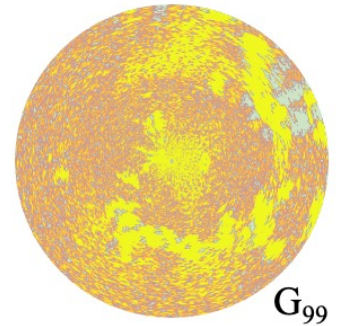
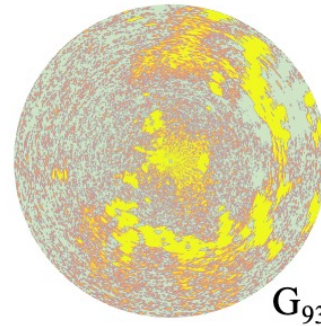
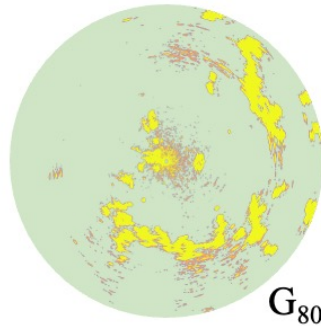
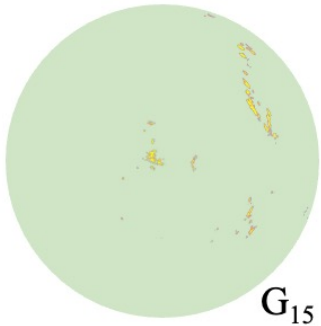
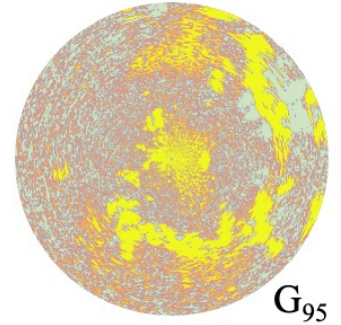
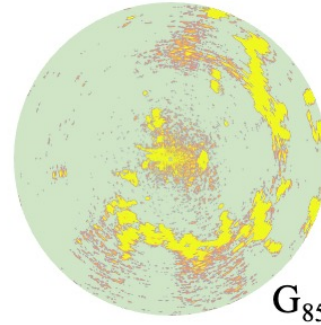
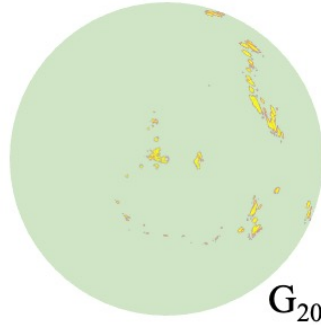
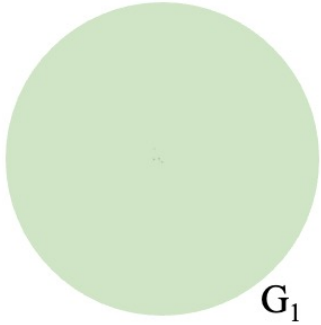
Radar Entropy – July '25

Noise Floor
Range

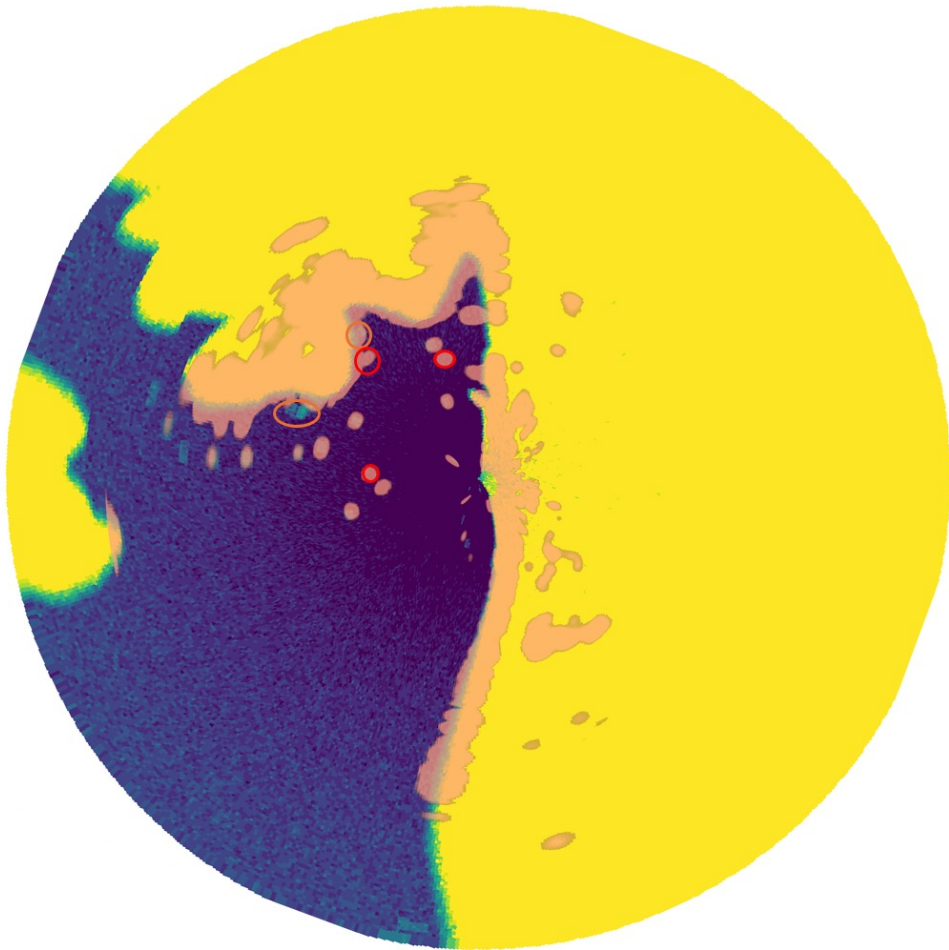
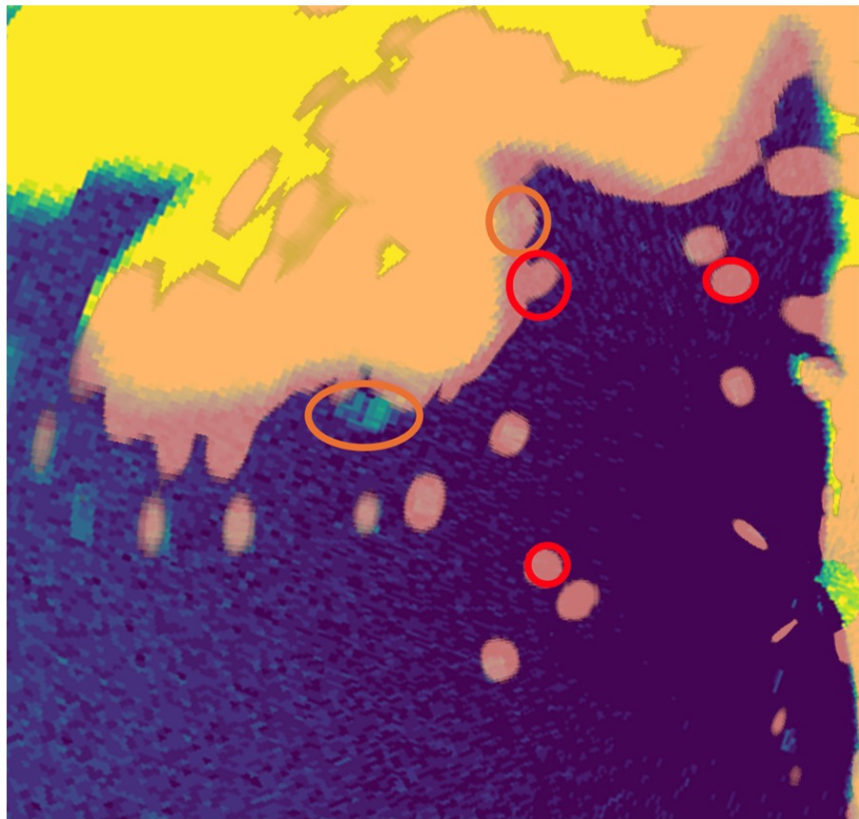
Target Range

Information
Zone

Oversaturation
Range



Radar Simulations – July '25



Radar Calibration Attempts – July '25



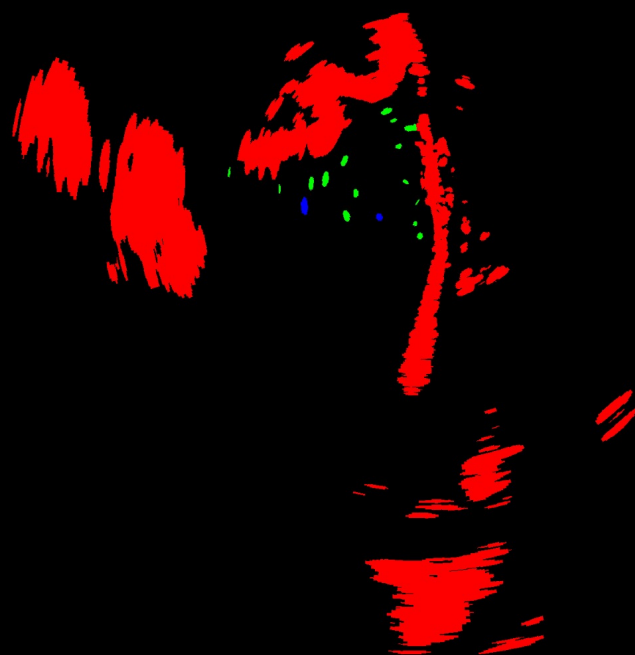
Object Detection via MAE ML – August '25

Initial Self-Supervised results with 3 labelled frames, thousands of unlabeled frames.

MAE Tracking(214px)



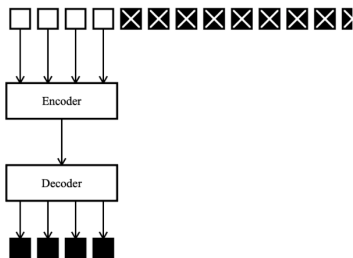
Manual Tracking(1735px)



Temporal Masked Auto Encoding – September ‘25

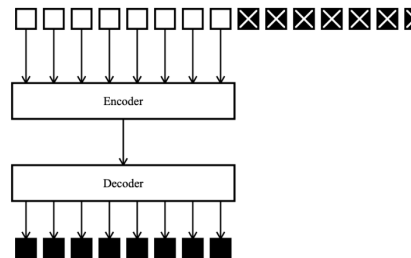
Fixed-Context (4 Frames)

Pads or truncates input to match its fixed architecture.



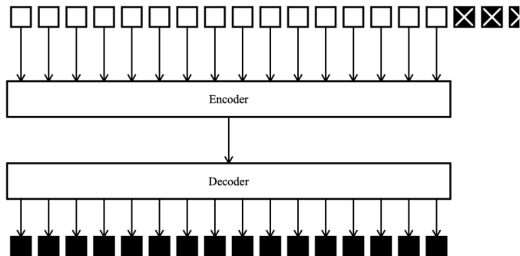
Fixed-Context (8 Frames)

Optimal only when input length is exactly 8.



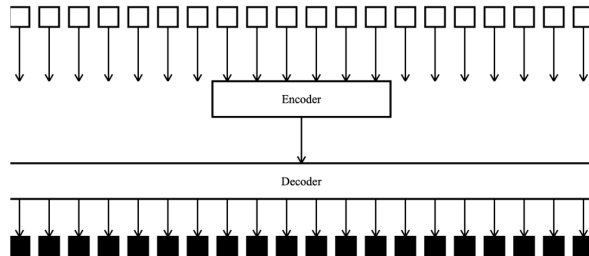
Fixed-Context (16 Frames)

Inefficient for short videos, loses info from long ones.

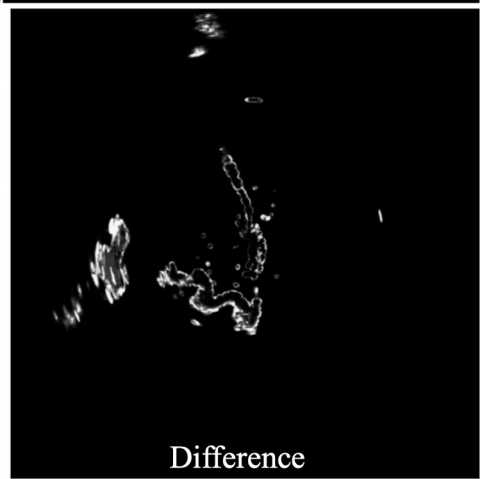
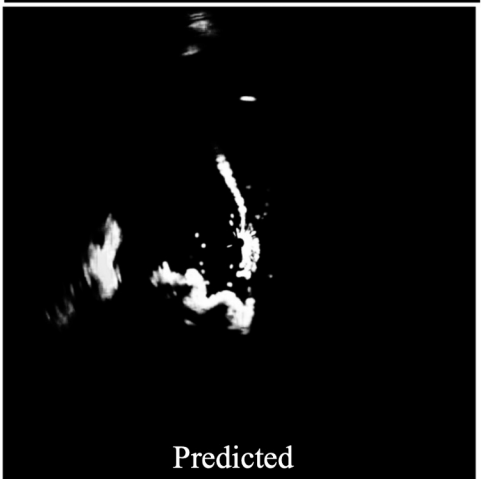
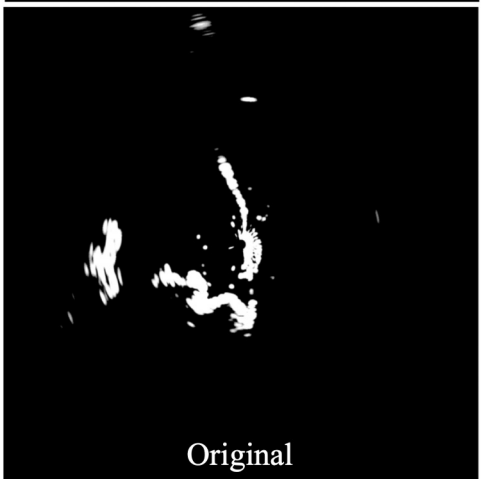
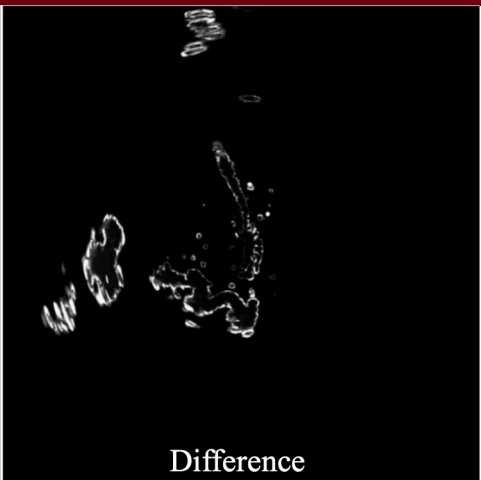
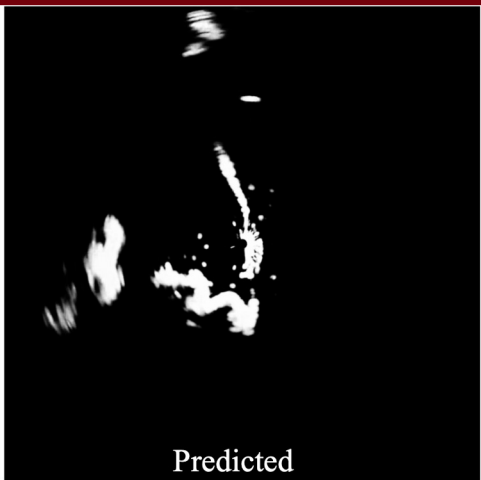
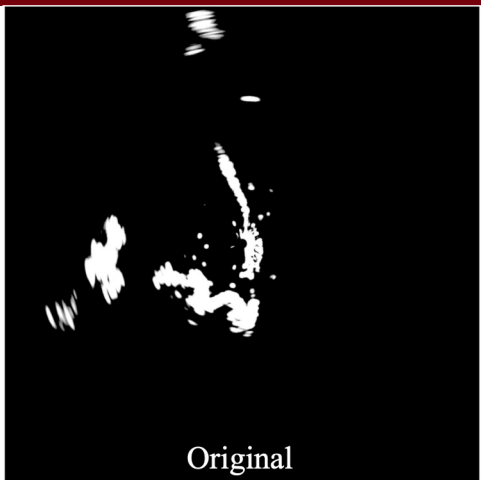


Variable-Context MAE

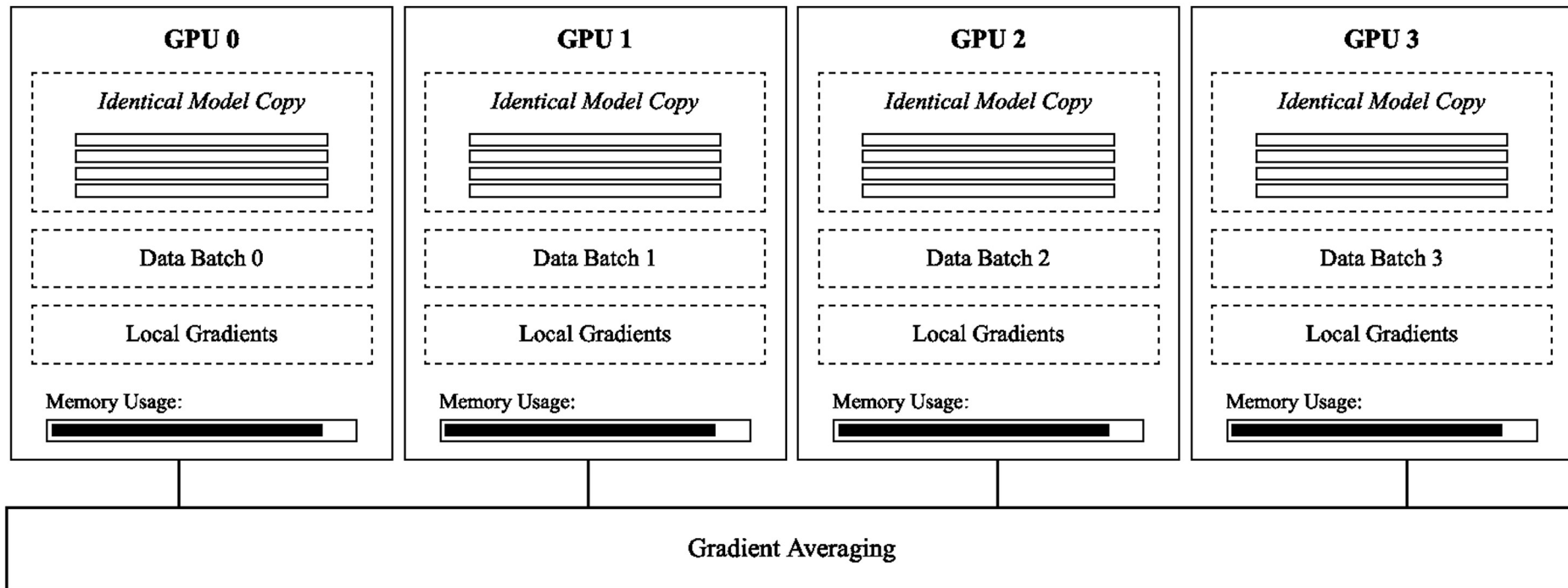
Dynamically adapts to any input length for optimal performance.



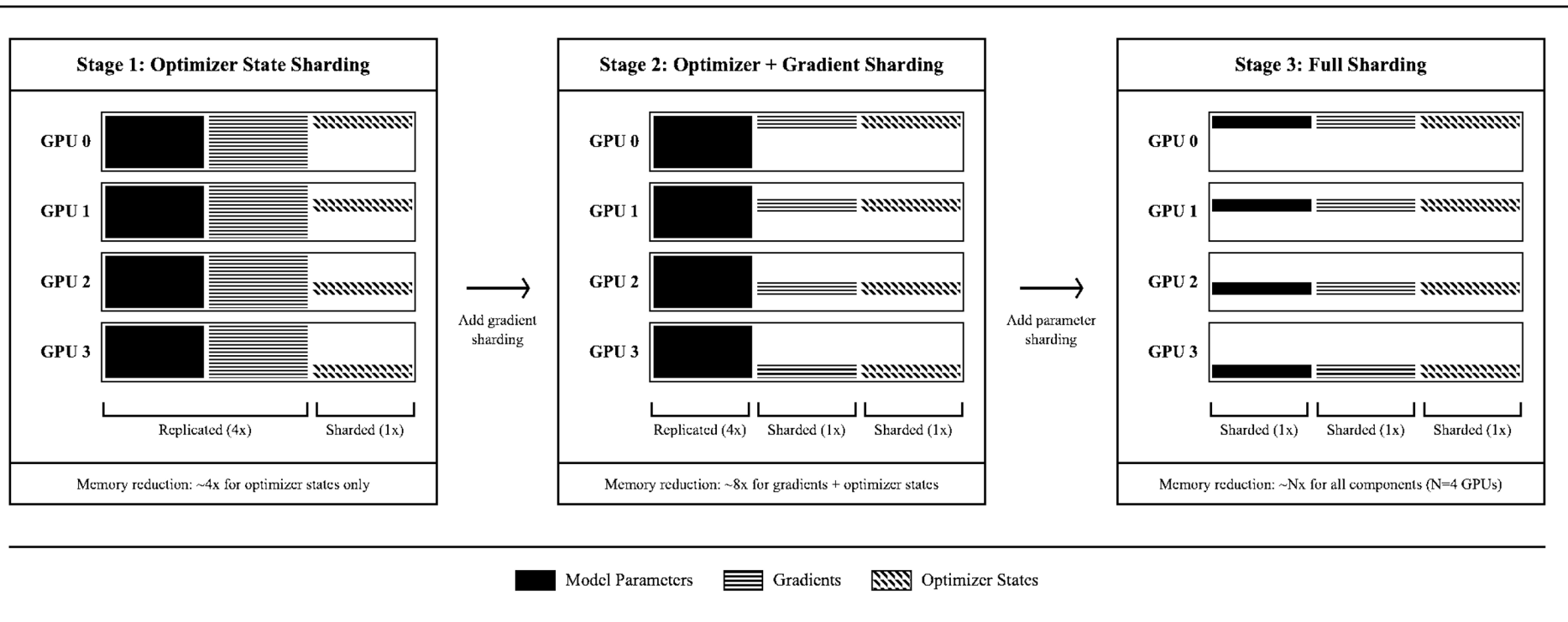
MAE Gain Prediction – September '25



GPU Sharding – September '25



DeepSpeed GPU Sharding – September ‘25



Mini Lit-Review for Radar Tracking – October '25

This is kinda the end-goal of what I would like to eventually have.

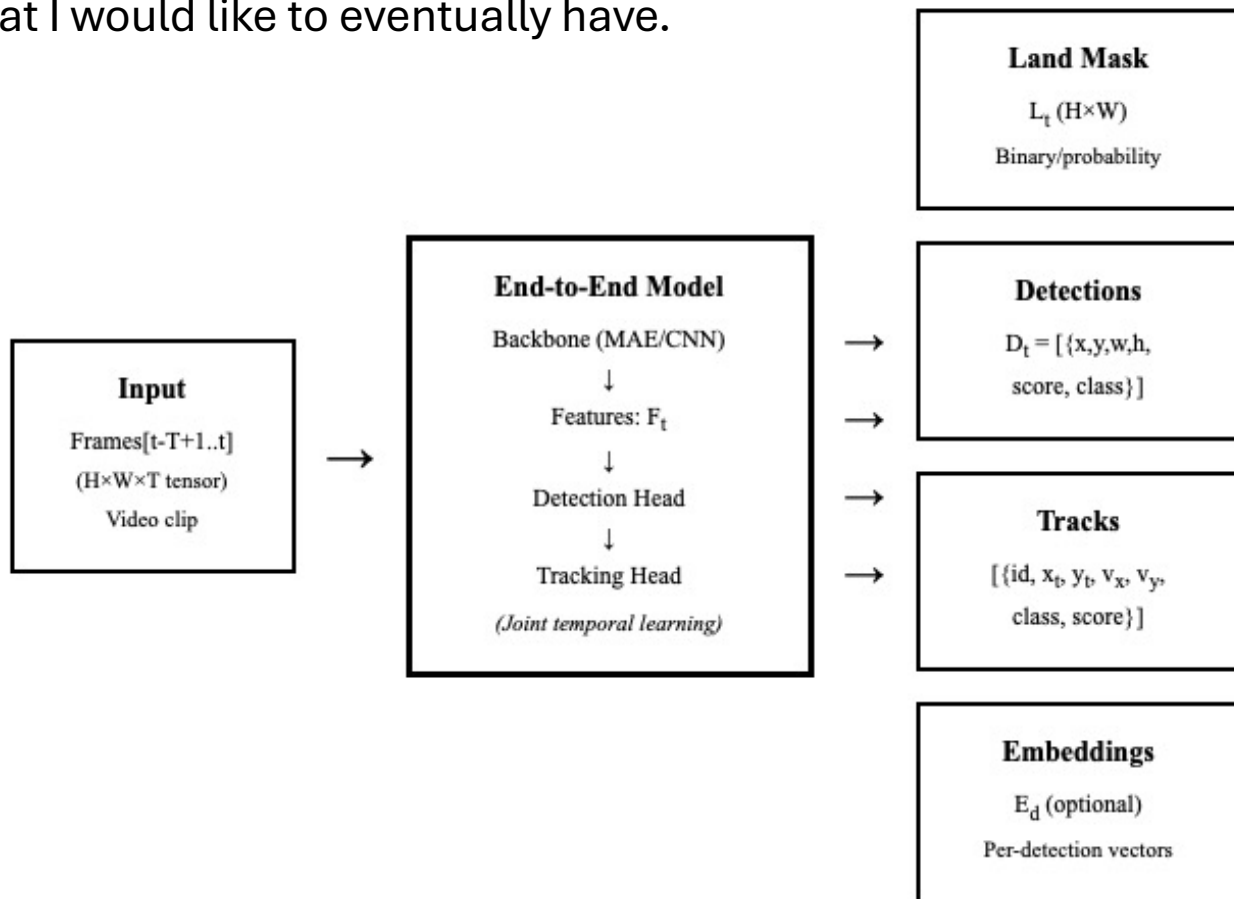
CenterTrack: 67.3% MOTA @ 22 FPS on MOT17, 89.4% MOTA on KITTI - online & real-time

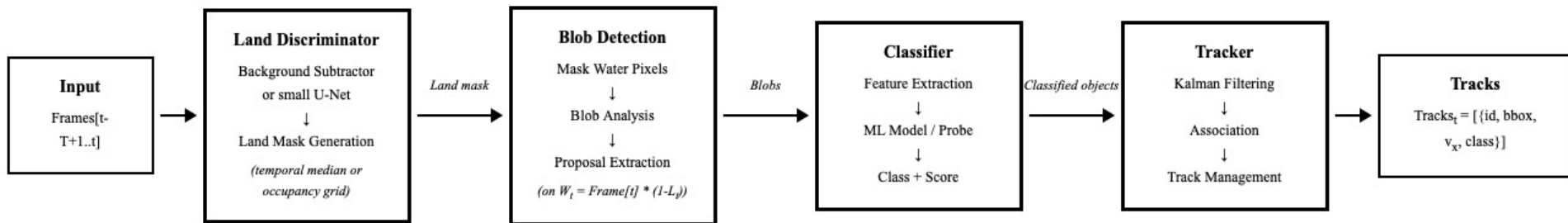
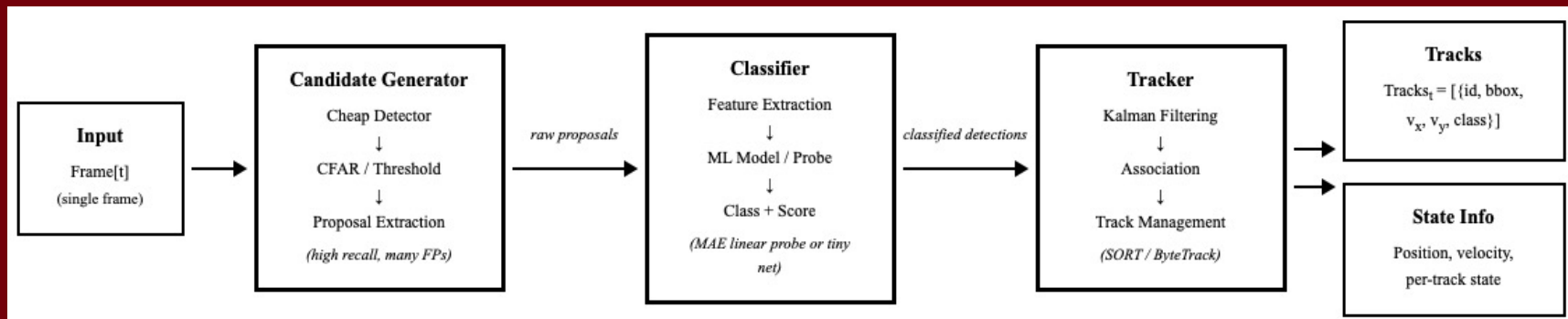
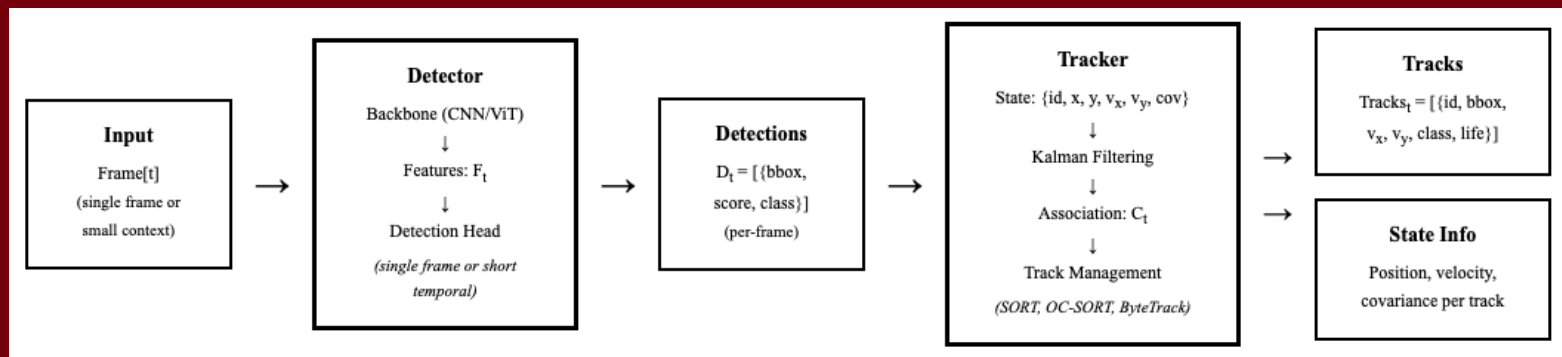
TransTrack: End-to-end MOT with unified detection+association using Transformer queries

TrackFormer: State-of-the-art on MOT17/MOT20 at publication (CVPR 2022)

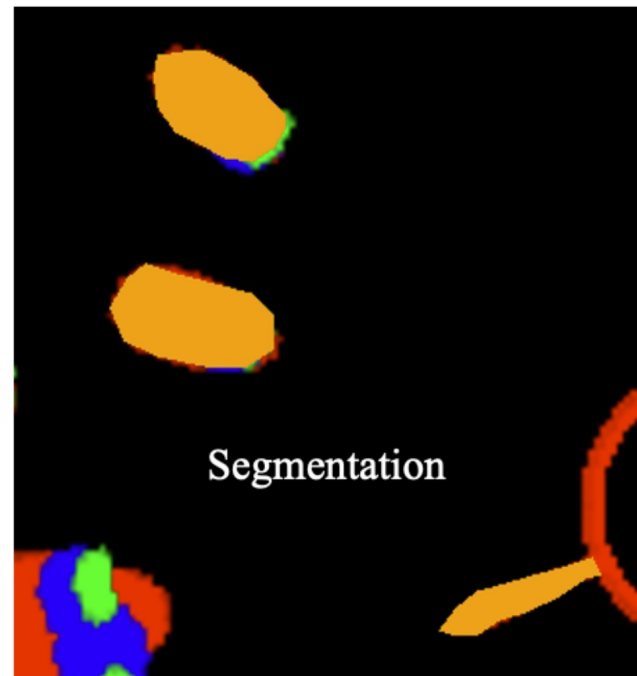
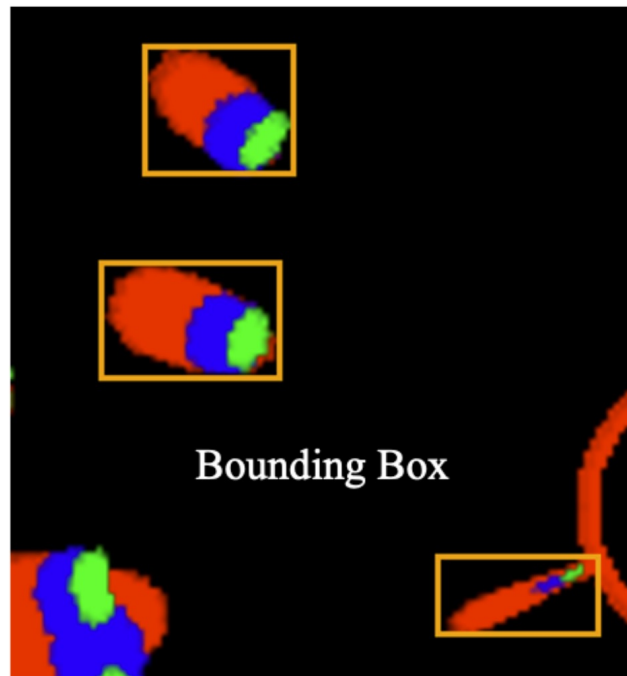
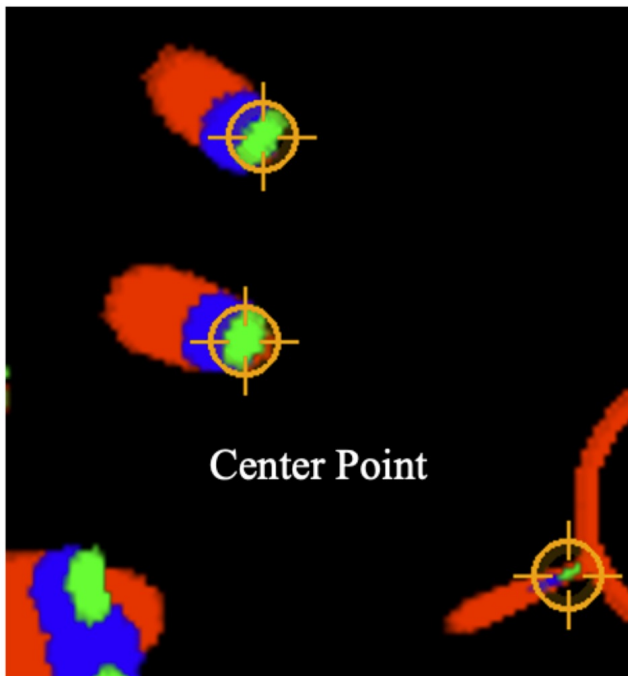
TransVOD++: 90.0% mAP on ImageNet-VID, ~30 FPS @ 83.7% mAP (Lite version)

CenterNet: 28.1% AP @ 142 FPS on COCO with multi-task heads





Lit-Review for Annotation Methods – October '25



Next Steps.

- **Oct 22-31:**

- Finish up NSF application
- Collect data in Hampton Roads area
- Fix radar issue

- **Nov 1-8:**

- Have working object tracking algorithm – target 70+% tracking of moving objects.

- **Later**

- Finish other applications
- Work on improving tracking on standard method as well as self-supervised.

Question for the audience.

Do you like the 'summary' of the project
before going into completed work?

Do you want a more technical presentation
in the future?

Questions?

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